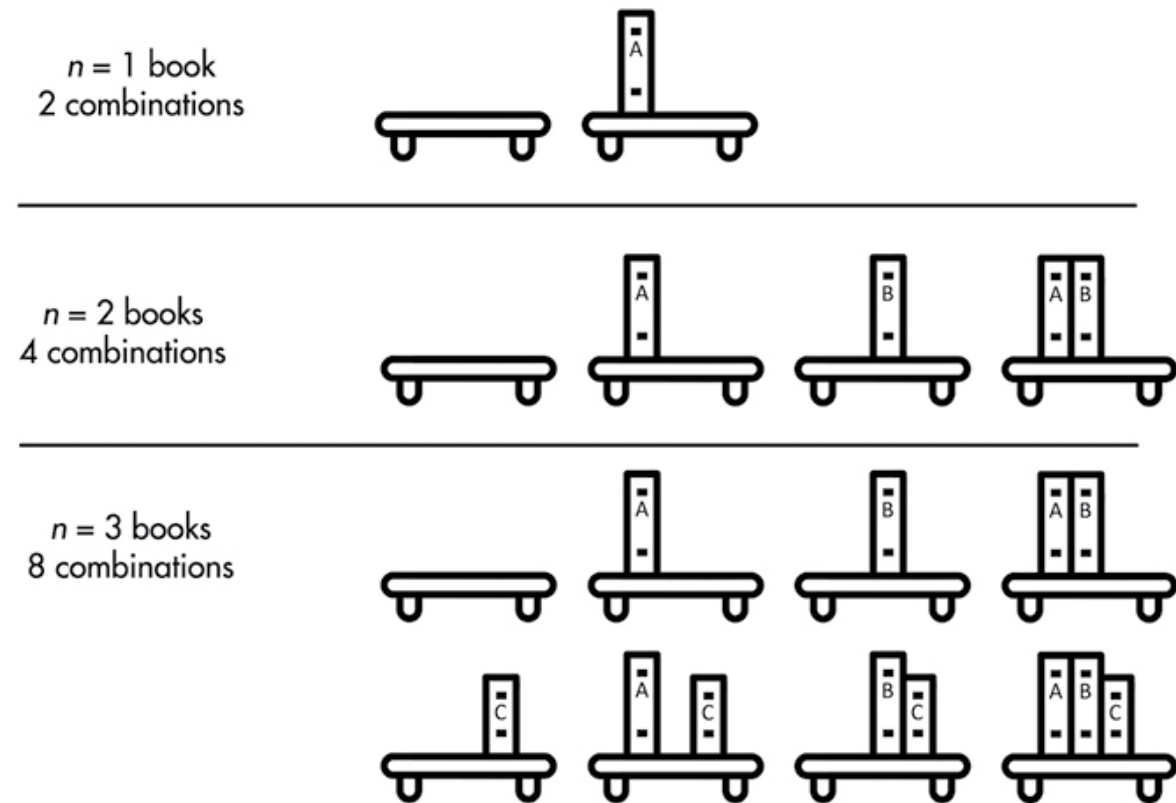




[Figure 13-1](#): Every combination of books on a bookshelf for one, two, or three books

The runtime for exponential tasks increases very quickly. Six books require 2^6 or 32 photos, but 32 books would include 2^{32} or more than 4.2 billion photos. $O(2^n)$, $O(3^n)$, $O(4^n)$, and so on are different orders but all have exponential time complexity.

$O(n!)$, Factorial Time

Taking a photo of the books on the shelf in every conceivable order is a factorial time operation. We call every possible order the *permutation* of n books. This results in $n!$, or n factorial, orderings. The *factorial* of a number is the multiplication product of all positive integers up to the number. For example, $3!$ is $3 \times 2 \times 1$, or 6. [Figure 13-2](#) shows every possible permutation of three books.